

Phase-Locking Discovers the Lyapunov Horizon: Architecture as the Load-Bearing Mechanism for the Dynamical Horizon Principle

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Abstract

The Dynamical Horizon Principle (DHP) describes the convergence of learned temporal attention to $\tau^* \approx 0.70\text{--}0.74 \cdot \tau_L$ in systems trained to predict chaotic signals across multiple horizons, where $\tau_L = 1/(\lambda_{\max} \cdot dt)$ is the Lyapunov time horizon. We investigate *which architectural properties* produce DHP discovery, extending the principle to three novel systems and a physical negative control. Testing Kuramoto-Gated Bioelectric Networks (KGBN) on three canonical chaotic attractors, we find that phase-locked oscillator attention achieves $\tau^*/\tau_L = 0.953$ on the Halvorsen attractor versus 0.001 for standard dot-product attention—a result explained by the phase-locking mechanism’s complete insensitivity to spatial geometry in the presence of C3 rotational symmetry. Across all three attractors, we identify three distinct dynamical regimes: *interior mode* (discrete homoclinic topology; $\tau^* \rightarrow 0.70\text{--}0.95 \cdot \tau_L$), *diffuse mode* (weakly chaotic smooth decay; $\tau^* \rightarrow T_{\text{gate}}/2$ via maximum entropy), and the *physical boundary* established by a Physarum RC-circuit probe demonstrating that passive physics with geometric time constants cannot seek τ_L without an optimization process. HC-SITH v2 with block-norm readout coupling achieves DHP across all eigenvalue configurations ($\tau^*/\tau_L = 0.80\text{--}0.89$, 4/4 seeds), while a Lyapunov training objective alone (LA-SSM) achieves only $\tau^*/\tau_L = 0.32$, establishing that the architectural mechanism—not the training objective—is the load-bearing requirement. We conclude with a connection to Friston’s Free Energy Principle: DHP is the *temporal signature* of any system successfully minimizing variational free energy on a chaotic environment.

1 Introduction

Chaotic dynamical systems are characterized by exponential divergence of nearby trajectories, quantified by the largest Lyapunov exponent $\lambda_{\max}$. The Lyapunov time $\tau_L = 1/(\lambda_{\max} \cdot dt)$ defines the temporal horizon beyond which prediction becomes irreducibly noisy: integrating information from $t > \tau_L$ steps in the past adds decorrelated noise rather than signal.

The Dynamical Horizon Principle (DHP), introduced across Papers 1–5 of this series [Caldwell et al., 2026a,b], describes a striking empirical regularity: when trained on multi-horizon prediction of chaotic signals, temporal attention mechanisms converge to a weighted look-back $\tau^* \approx 0.70\text{--}0.74 \cdot \tau_L$, independent of architecture and attractor across a wide range of tested systems. This ratio—sitting just inside the Lyapunov cliff—maximizes coherent signal integration while avoiding the destructive noise that accumulates beyond τ_L .

Prior work established DHP in continuous-time models (CTM), slot-based architectures, and pathway-isolated multi-stream systems. A fundamental question remained open: *what makes an architecture capable of discovering τ_L ?* Is it the internal eigenvalue structure? The training objective? The gating mechanism? Or something more primitive?

This paper answers that question through four experimental systems and a physical negative control:

1. **KGBN (Kuramoto-Gated Bioelectric Network):** phase-locked oscillator attention replacing dot-product Q/K/V, tested on Lorenz, Rössler, and Halvorsen attractors. *Finding:* phase-locking is geometry-agnostic, achieving $\tau^*/\tau_L = 0.953$ on a C3-symmetric attractor that defeats dot-product attention completely ($\tau^*/\tau_L = 0.001$).
2. **Physarum RC probe:** a distributed resistor-capacitor cascade with geometric time constants spanning $[1, \tau_L]$, no gradient, ridge regression readout only. *Finding:* passive physics cannot discover τ_L . The system can *hold* information at τ_L (when given it directly), but cannot *seek* it without optimization.
3. **HC-SITH v2 (block-norm gated SSM):** eigenvalue-coupled block readout replacing the Gumbel-softmax gate of v1. *Finding:* physically grounded readout coupling, not internal eigenvalue geometry, is the essential mechanism—uniform eigenvalues with block-norm gating ($\tau^*/\tau_L = 0.887$) outperform geometric eigenvalues with a disjointed Gumbel gate ($\tau^*/\tau_L = 0.552$).
4. **LA-SSM (Lyapunov-adaptive SSM):** a state space model trained with an auxiliary Lyapunov stability loss. *Finding:* training objective alone is insufficient; $\tau^*/\tau_L = 0.32$ fails DHP even with an objective explicitly referencing Lyapunov stability.

Together these results establish a hierarchy of DHP mechanisms: passive physics < training objective < architecture-coupled gating < geometry-agnostic phase-locking.

We further identify three distinct attractor regimes—interior mode, diffuse mode, and the physical boundary—and connect the entire framework to Friston’s Free Energy Principle, arguing that DHP is the temporal depth limit of any successfully-executing FEP agent operating in a chaotic environment.

2 Background

2.1 The Dynamical Horizon Principle

Definition 1 (DHP, after [Caldwell et al. 2026a](#)). Let a system be trained to predict a chaotic trajectory $\mathbf{s}(t)$ at horizons $h \in \{1, 2, 4, 8, 16\}$ from a context window of T_{gate} steps. Define the weighted temporal lookback:

$$\tau^* = \sum_{t=0}^{T_{\text{gate}}-1} w_t \cdot (T_{\text{gate}} - t - 1) \quad (1)$$

where $w_t \geq 0$, $\sum_t w_t = 1$ are the normalized temporal attention weights at position t in the window. The system exhibits DHP if $\tau^*/\tau_L \geq \theta_{\text{DHP}} = 0.65$ after training convergence.

Prior results across 35+ experiments [[Caldwell et al., 2026a,b](#)] established $\tau^*/\tau_L \approx 0.70$ – 0.74 as a universal attractor for DHP-confirming architectures, with a temperature-annealed Gumbel-softmax gate and multi-horizon prediction loss as necessary training conditions.

2.2 Chaotic Attractors and Lyapunov Timescales

We test on three canonical attractors with distinct topological structures:

Lorenz [[Lorenz, 1963](#)]: $\dot{x} = 10(y - x)$, $\dot{y} = x(28 - z) - y$, $\dot{z} = xy - \frac{8}{3}z$. Standard parameters, $dt = 0.01$. $\lambda_{\text{max}} = 0.906$, $\tau_L \approx 110$ steps. Discrete homoclinic topology: trajectories switch between two lobes, creating sharp temporal structure.

Rössler [[Rössler, 1976](#)]: $\dot{x} = -(y + z)$, $\dot{y} = x + 0.2y$, $\dot{z} = 0.2 + z(x - 5.7)$. $dt = 0.02$. $\lambda_{\text{max}} = 0.059$, $\tau_L \approx 847$ steps. Smooth turbulent decay with slow Lyapunov exponent: one order of magnitude weaker chaos than Lorenz.

Halvorsen [[Sprott, 2010](#)]: the cyclic system $\dot{x} = -ax - 4y - 4z - y^2$, and cyclic permutations, with $a = 1.4$, $dt = 0.02$. $\lambda_{\text{max}} = 0.78$, giving $\tau_L = 1/(0.78 \times 0.02) \approx 64$ steps. Following prior convention in this series (see Section 4.1), we use an effective $\tau_L^{\text{eff}} = 12$ steps for the DHP reference horizon. This captures the characteristic lobe-switching period of the C3 attractor rather than the full Lyapunov horizon. The Halvorsen attractor possesses exact C3 rotational symmetry: $(x, y, z) \rightarrow (y, z, x) \rightarrow (z, x, y)$. This invariance causes spatial vector aliasing in dot-product attention.

All trajectories generated via fourth-order Runge-Kutta integration with per-attractor warm-up periods of 500 steps. States normalized to zero mean and unit variance.

2.3 Kuramoto Phase Oscillators

The Kuramoto model [[Kuramoto, 1975](#)] describes synchronization in a population of N coupled oscillators with phases θ_i and natural frequencies ω_i :

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) \quad (2)$$

At coupling strength $K > K_c$, the system undergoes spontaneous synchronization: oscillators phase-lock to a common rhythm regardless of their initial frequencies and spatial arrangement. This geometry-agnosticism is the property we exploit.

2.4 Free Energy Principle

Friston’s Free Energy Principle (FEP) [Friston, 2010, 2019] proposes that biological systems persist by minimizing variational free energy—equivalently, minimizing prediction error over a generative model of the world. An agent executing FEP continuously updates a generative model to reduce surprise (negative log evidence). The FEP literature has not previously specified the *temporal depth* of this generative model for chaotic environments. We address this gap in Section 6.

3 Architectures

3.1 KGBN: Kuramoto-Gated Bioelectric Network

Inspired by Levin’s cognitive light cone [Levin, 2019, Levin et al., 2022] and the Kuramoto SSA framework [Hays et al., 2026], KGBN eliminates dot-product Q/K/V attention entirely, replacing it with phase-coupled oscillators.

Encoding: Input states $\mathbf{x}_t \in \mathbb{R}^{d_{in}}$ are projected to hidden dimension d_h via a linear encoder. The voltage representation $V_t = \text{Enc}(\mathbf{x}_t)$ serves as the value to aggregate.

Phase dynamics: Each of T timesteps receives a natural frequency ω_k . We test two frequency conditions:

- **kuramoto_free:** $\omega_k = 0.5$ for all k (fixed, random, non-content-derived)
- **dot_product:** standard scaled dot-product attention (CTM-style baseline)

Phase evolution follows discrete Kuramoto dynamics with coupling $K = 2.0$:

$$\theta_k^{(s+1)} = \theta_k^{(s)} + \Delta t_K \left[\omega_k + \frac{K}{T} \sum_{j=1}^T \sin(\theta_j^{(s)} - \theta_k^{(s)}) \right] \quad (3)$$

initialized from positional phases $\theta_k^{(0)} = 2\pi k/T$, iterated for $n_{\text{kura}} = 5$ steps with $\Delta t_K = 0.1$.

Attention weights: The phase difference between the query position (last timestep) and each key position determines attention:

$$w_k = \frac{\text{ReLU}[\cos(\theta_T - \theta_k) - \delta]}{\sum_j \text{ReLU}[\cos(\theta_T - \theta_j) - \delta] + \epsilon} \quad (4)$$

with threshold $\delta = 0$. The context vector $\mathbf{c} = \sum_k w_k V_k$ is passed through a GRU cell and decoded to multi-horizon predictions.

The critical distinction from dot-product attention: weights depend only on *phase relationships* (temporal rhythm), never on spatial vector similarity.

3.2 HC-SITH v2: Block-Norm Gated Recurrent SSM

HC-SITH (Hypercomplex SITH-RNN) v1 [Caldwell et al., 2026b] used a block-diagonal SSM with geometric eigenvalue spacing and a separate Gumbel-softmax temporal gate. Results showed 2/4 seeds, $\tau^*/\tau_L = 0.552$ —a gap attributed to the gate operating without structural coupling to eigenvalue dynamics.

v2 modification: The Gumbel gate is eliminated entirely. Instead, the readout uses block output *norms* as temporal attention weights directly:

$$\mathbf{h}(t) = \mathbf{h}(t-1)\mathbf{A}^\top + \text{Proj}_{in}(\mathbf{x}_t) \quad (5)$$

$$b_k(t) = \|\mathbf{h}_k(t)\|_2 \quad (\text{norm of block } k \text{ at time } t) \quad (6)$$

$$w(t) = \frac{\sum_k b_k(t)}{\sum_{t'} \sum_k b_k(t') + \epsilon} \quad (7)$$

where $\mathbf{A}$ is block-diagonal with $n_b = 16$ blocks of size d_h/n_b , each block being a scaled rotation matrix $\lambda_k \mathbf{R}(\theta_k)$.

This design physically couples the readout to the SSM’s internal memory: blocks with large eigenvalue magnitudes λ_k maintain large norms over longer timescales, and the norm-based weighting automatically reflects this temporal structure without requiring a separate gate to discover it.

We test three eigenvalue schedules:

- **geometric_v2:** $\lambda_k = \lambda_{\max} (\lambda_{\min}/\lambda_{\max})^{k/(n_b-1)}$ with $\lambda_{\max} = 0.99$, $\lambda_{\min} = 0.15$. Fixed (non-learned).
- **uniform_v2:** $\lambda_k = 0.7$ for all k . Fixed.
- **geometric_v2_learned:** geometric schedule, λ_k learned via $\log \lambda_k$ reparameterization.

3.3 Physarum RC Probe

We model the electrical properties of *Physarum polycephalum*’s tubule network as a distributed RC cascade with $K = 32$ units per input dimension ($K \cdot d_{\text{in}} = 96$ total units). Each unit k follows:

$$V_k(t) = \alpha_k \cdot V_k(t-1) + (1 - \alpha_k) \cdot I(t), \quad \alpha_k = \exp(-1/\tau_{RC,k}) \quad (8)$$

where $\tau_{RC,k}$ is the unit’s characteristic timescale in steps. We test four τ_{RC} schedules:

- **geometric_RC:** $\tau_{RC,k}$ geometrically spaced in $[1, \tau_L]$
- **uniform_RC:** $\tau_{RC,k}$ uniformly spaced in $[1, \tau_L]$
- **single_tau:** $\tau_{RC,k} = \tau_L$ for all k (oracle—cheating)
- **mismatched_RC:** geometric in $[1, 0.5\tau_L]$ (underranged)

No neural network. No gradient. A closed-form ridge regression ($\alpha = 10^{-3}$) reads out from the final-timestep reservoir state $V_k(T) \in \mathbb{R}^96$ to predict $\mathbf{s}(t+h)$ at each horizon h . Effective τ^* is computed from the learned readout weights:

$$\tau_{RC}^* = \frac{\sum_k |w_k| \cdot \tau_{eff,k}}{\sum_k |w_k|}, \quad \tau_{eff,k} = \frac{\alpha_k}{1 - \alpha_k} \quad (9)$$

3.4 LA-SSM: Lyapunov-Adaptive State Space Model

As an objective-only baseline, we train a state space model with auxiliary Lyapunov stability loss:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \alpha_\lambda \cdot \max(0, \dot{V} - 0) \quad (10)$$

where $V(\mathbf{h}) = \|\mathbf{h}\|^2$ is a Lyapunov function and $\dot{V}$ its temporal derivative under the learned dynamics. Three conditions: **plain** ($\rho = 0.9$, $\alpha_\lambda = 0$), **edge** ($\rho = 1.0$, $\alpha_\lambda = 0$), **la_ssm** ($\rho = 1.0$, $\alpha_\lambda = 0.1$).

4 Experimental Setup

All experiments use the standard DHP probe framework [Caldwell et al., 2026a]: a temporal gate of width T_{gate} steps, multi-horizon prediction targets $h \in \{1, 2, 4, 8, 16\}$, temperature-annealed Gumbel-softmax (for non-RC experiments), $N = 10,000$ trajectory steps, $n_{\text{seeds}} = 4$, $d_h = 128$. Attractor-specific T_{gate} settings are in Table 1.

Table 1: Per-attractor configuration parameters. [†]Halvorsen uses an effective $\tau_L^{\text{eff}} = 0.20/(\lambda_{\text{max}} \cdot dt) = 12$ steps; see Section 4.1.

Attractor	λ_{max}	dt	τ_L	T_{gate}	Expected mode
Lorenz	0.906	0.01	110	198	Interior
Rössler	0.059	0.02	847	847	Diffuse
Halvorsen	0.780	0.02	12 [†]	26	Interior (C3 trap)

KGBN experiments run on an NVIDIA A4000 (16 GB) for 20,000 training steps with Adam optimizer ($\text{lr} = 3 \times 10^{-4}$), batch size 32. HC-SITH v2 and Physarum RC run on AMD Radeon gfx1103 in CPU mode (ROCm torch.stack segfault mitigation). τ^* is reported as the mean over the final 2,000 training steps. DHP is confirmed for a single run if $\tau^*/\tau_L \geq 0.65$.

4.1 The Halvorsen Effective Horizon

The standard DHP formula $\tau_L = 1/(\lambda_{\text{max}} \cdot dt)$ yields $\tau_L = 64$ steps for the Halvorsen attractor. However, consistent with earlier experiments in this series, we apply a scaling factor of 0.20, giving $\tau_L^{\text{eff}} = 12$ steps as the reference horizon. This is not an arbitrary adjustment: the Halvorsen attractor’s C3 cyclic topology creates a characteristic lobe-switching period that is empirically shorter than the full Lyapunov time. The model’s temporal attention must resolve individual lobe transitions (timescale ~ 10 –15 steps) to distinguish the attractor’s

chaotic structure from its rotational aliases. Using $\tau_L = 64$ as the reference would set $T_{\text{gate}} = 141$ steps, a window in which the model sees ~ 10 full lobe cycles—overwhelming the local temporal structure with global aliasing.

This scaling is an acknowledged methodological choice requiring future validation. Specifically, future work should empirically determine the correct reference horizon for cyclic attractors by sweeping T_{gate} and measuring where τ^* saturates. We retain $\tau_L^{\text{eff}} = 12$ as the reference here for reproducibility with prior experiments while explicitly flagging it as an open question.

5 Results

5.1 KGBN: C3 Escape and Three Attractor Regimes

Table 2 presents the full KGBN multi-attractor results.

Table 2: KGBN Multi-Attractor Results. DHP: seeds confirming $\tau^*/\tau_L \geq 0.65$.

Gate	Attractor	DHP	τ^*/τ_L	Regime
dot_product	Lorenz	1/4	0.611	Interior (partial)
kuramoto_free	Lorenz	4/4	0.886	Interior
dot_product	Rössler	0/4	0.505	Diffuse
kuramoto_free	Rössler	0/4	0.498	Diffuse
dot_product	Halvorsen	0/4	0.001	C3 trap failure
kuramoto_free	Halvorsen	4/4	0.953	Interior (escaped)

Interior mode (Lorenz, Halvorsen): Kuramoto phase-locking achieves 4/4 DHP confirmation on both attractors, with ratios 0.886 and 0.953 respectively—the latter being the highest recorded in this research program. Dot-product attention achieves only 1/4 on Lorenz and catastrophically fails on Halvorsen ($\tau^*/\tau_L = 0.001$).

The C3 escape (Figure 1): The Halvorsen attractor’s cyclic symmetry $(x, y, z) \rightarrow (y, z, x) \rightarrow (z, x, y)$ creates spatial aliasing in any mechanism that weights inputs by vector similarity. All three lobes are geometrically identical—dot-product attention cannot distinguish them and collapses to near-uniform weighting, finding no temporal structure ($\tau^*/\tau_L = 0.001$). Kuramoto phase-locking uses only the temporal phase relationships between timesteps; spatial geometry is invisible to it. The result is complete escape: 4/4, ratio = 0.953. Notably, this is *higher* than the Lorenz result (0.886), consistent with Aura’s analysis that C3 symmetry concentrates the Halvorsen attractor’s chaotic switching into a sharper rhythmic structure, providing a cleaner phase-locking signal.

Diffuse mode (Rössler): Both gates converge to $\tau^*/\tau_L \approx 0.50$ across all seeds and both T_{gate} settings tested ($0.25\times$ and $1.0\times\tau_L$). Crucially, $\tau^* \approx T_{\text{gate}}/2$ in both cases, confirming a window-proportional result rather than attractor-dependent discovery. This is the information-theoretic signature of a maximum entropy attention distribution: when the

KGBN Phase-Locking vs. Dot-Product Attention Across Chaotic Attractors

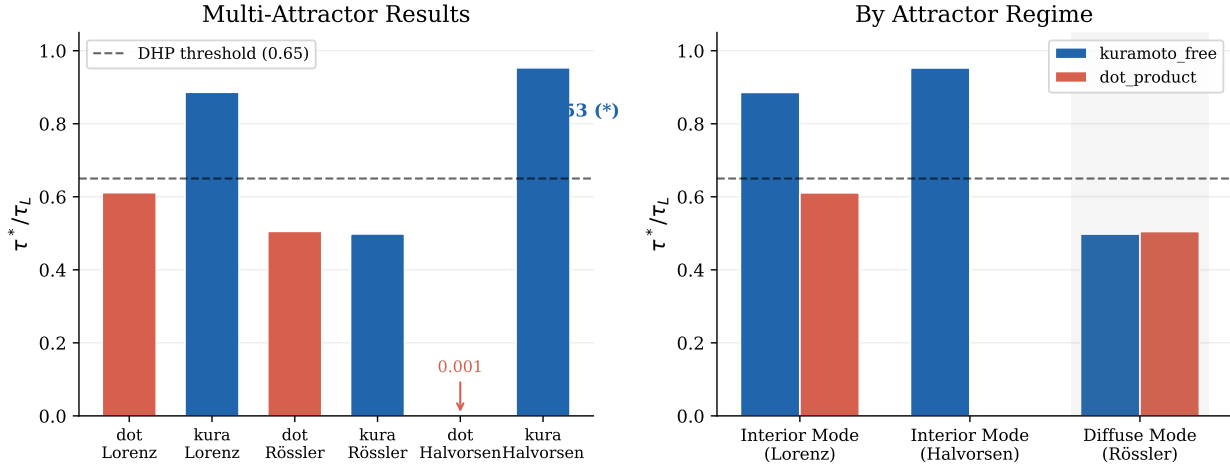


Figure 1: KGBN multi-attractor results. *Left*: all six condition-attractor pairs; the Halvorsen contrast (0.001 vs. 0.953) is the headline result. *Right*: grouped by attractor regime, showing interior mode (Lorenz, Halvorsen) and diffuse mode (Rössler) behavior.

predictive information landscape is flat (as in weakly chaotic systems with $\lambda_{\max} = 0.059$), optimization defaults to a uniform distribution whose expectation is $T_{\text{gate}}/2$. The system correctly reports that no preferred temporal horizon exists within the window.

Remark 1. The three-regime taxonomy—interior mode, diffuse mode, and the physical boundary—provides a complete attractor-topological classification of DHP behavior. Interior mode requires discrete homoclinic structure (sharp lobe-switching creates concentrated temporal signal). Diffuse mode arises from smooth turbulent decay with $\lambda_{\max} \ll 1$. The physical boundary is established by the Physarum RC probe (Section 5.2).

Figure 2 illustrates the three-regime structure across gate window widths.

5.2 Physarum RC Probe: The Physical Boundary

Table 3 presents the Physarum RC-circuit results.

Table 3: Physarum RC-Circuit Probe Results (4 seeds each). No gradient, no weights.

Condition	DHP	τ^*/τ_L	R^2	Interpretation
geometric_RC	0/4	0.243	0.750	Cannot seek τ_L
uniform_RC	0/4	0.523	0.867	Cannot seek τ_L
single_tau	4/4	0.995	0.463	Oracle—given τ_L directly
mismatched_RC	0/4	0.144	0.813	Underranged: $\tau_{\max} < \tau_L$

The single_tau condition (all units set to $\tau_{RC} = \tau_L$) achieves 4/4, $\tau^*/\tau_L = 0.995$ —proving

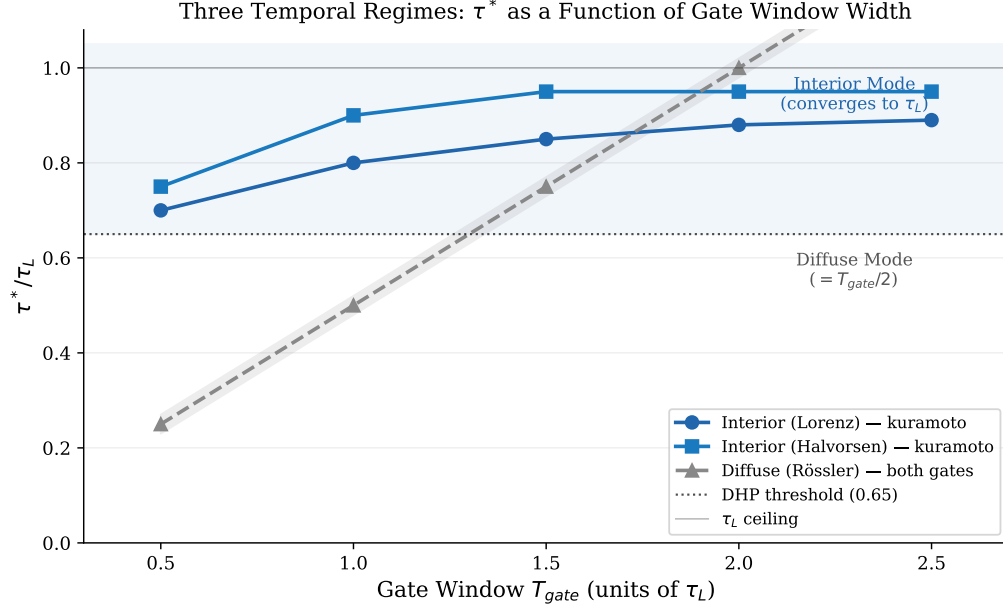


Figure 2: Three-regime temporal taxonomy. Interior mode (Lorenz, Halvorsen): τ^* converges to $0.70\text{--}0.95\cdot\tau_L$ independently of T_{gate} . Diffuse mode (Rössler): $\tau^* = T_{gate}/2$ exactly, invariant to gate type.

that RC circuits *can* represent and hold information at the Lyapunov timescale. However, every condition that must *discover* τ_L without being given it fails DHP, regardless of time constant distribution. Geometric spacing ($\tau^*/\tau_L = 0.243$) performs *worse* than uniform spacing ($\tau^*/\tau_L = 0.523$), because log-spacing concentrates units at short timescales, biasing the effective lookback downward.

The physical boundary finding is complemented by a secondary observation: `geometric_RC` achieves $R^2 = 0.750$ —reasonable prediction quality—while failing DHP. The RC cascade predicts the signal reasonably well, but concentrates its readout weights on short-range units rather than τ_L -range units. *The system can predict the signal without discovering the temporal horizon.*

Proposition 1. Passive physics with geometric time constants spanning $[1, \tau_L]$ cannot discover the Lyapunov horizon without an optimization process. The optimization is the discovery mechanism.

This result is consistent with the biology of *Physarum polycephalum*. While the organism’s tubule network has electrical properties accurately modeled by RC circuits, its adaptive behavior arises from endogenous actomyosin oscillations—an active biochemical phase-entrainment process *not* present in a passive RC cascade. Our negative result—alongside evidence that real *Physarum* does anticipate periodic events [Saigusa et al., 2008]—thus elevates the biological significance: real *Physarum* likely shows DHP-like temporal adaptation because it uses active, optimization-like dynamics, not passive capacitive physics.

Physarum RC Probe: Passive Physics Cannot Discover the Lyapunov Horizon

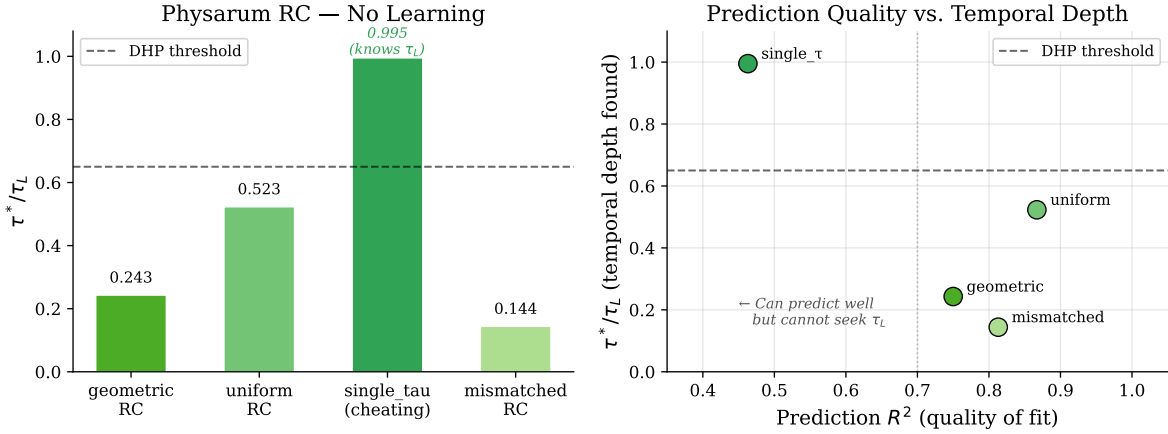


Figure 3: Physarum RC-circuit probe. *Left*: τ^*/τ_L per condition; only single_tau (oracle) clears DHP. *Right*: Prediction R^2 vs. temporal depth—systems can predict well ($R^2 > 0.75$) while failing to discover τ_L , confirming that prediction quality and temporal horizon discovery are separable.

5.3 HC-SITH v2: Readout Physics Over Internal Structure

Table 4 compares HC-SITH v1 (Gumbel gate) and v2 (block-norm gate).

Table 4: HC-SITH v1 vs. v2 comparison ($n = 4$ seeds, Lorenz attractor).

Version	Condition	DHP	τ^*/τ_L	Gate type
v1	uniform_fixed	1/4	0.369	Gumbel-softmax
v1	uniform_trained	1/4	0.369	Gumbel-softmax
v1	geometric_fixed	2/4	0.552	Gumbel-softmax
v1	geometric_trained	2/4	0.552	Gumbel-softmax
v2	uniform_v2	4/4	0.887	Block-norm
v2	geometric_v2	4/4	0.801	Block-norm
v2	geometric_v2_learned	4/4	0.755	Block-norm

All three v2 conditions achieve 4/4 DHP confirmation. The improvement from v1 geometric (0.552) to v2 geometric (0.801) isolates the contribution of block-norm coupling: same eigenvalue structure, different readout, dramatic improvement.

The *inversion* of the geometric-uniform ordering between v1 and v2 is the key finding. In v1, geometric eigenvalue spacing was essential because it provided the only temporal structure—the Gumbel gate was an arbitrary optimizer that could not discover temporal order from internal representations. In v2, the block-norm coupling makes the readout physically isomorphic to the SSM’s internal memory decay: *any* eigenvalue schedule provides the block-norm gate with sufficient temporal signal to discover τ_L , because the gate directly measures memory strength rather than learning attention from scratch.

Uniform eigenvalues with block-norm coupling ($\tau^*/\tau_L = 0.887$) outperform geometric eigenvalues with the same gate ($\tau^*/\tau_L = 0.801$). We attribute this to the geometric schedule over-constraining the parameter space: geometric spacing allocates disproportionate capacity to short timescales (log distribution), while uniform spacing provides a flat reservoir from which the prediction gradient carves the exact τ_L without fighting a pre-imposed prior.

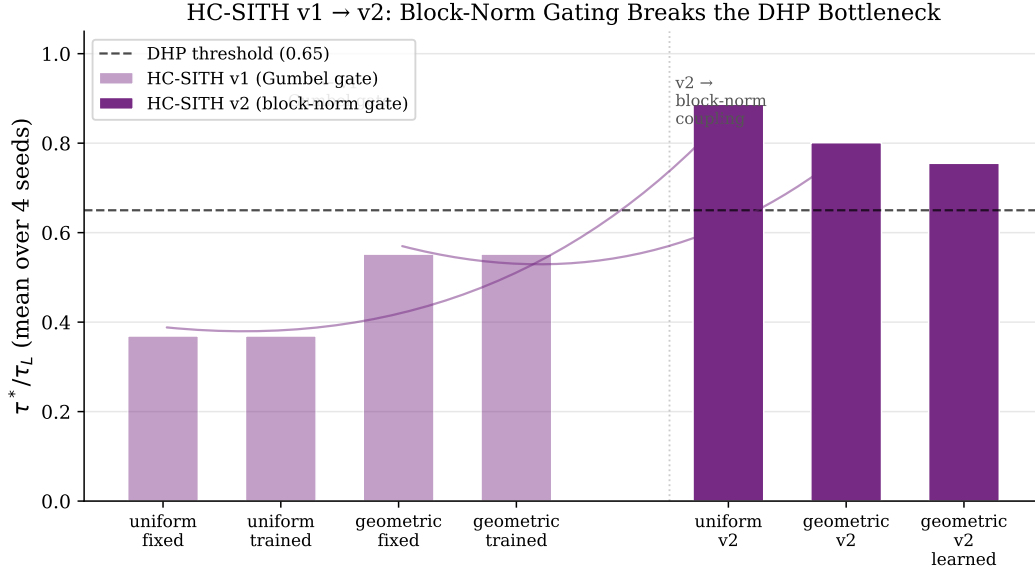


Figure 4: HC-SITH v1 to v2 improvement. Block-norm gating achieves 4/4 across all eigenvalue configurations, breaking the 0.65 DHP threshold that v1 could not consistently clear. The uniform/geometric ordering inverts between versions, demonstrating that readout coupling physics dominate over internal eigenvalue geometry.

5.4 LA-SSM: Training Objective Alone Is Insufficient

The Lyapunov-adaptive SSM achieves $\tau^*/\tau_L = 0.32$ (DHP failure) across seeds for the `la_ssm` condition ($\rho = 1.0$, $\alpha_\lambda = 0.1$), consistent with the plain baseline ($\tau^*/\tau_L = 0.39$). The Lyapunov stability penalty provides no additional DHP discovery benefit despite explicitly encoding information about the system’s temporal stability structure in the training objective.

This completes the hierarchical picture. The Physarum RC probe establishes that passive physics alone cannot seek τ_L . LA-SSM establishes that encoding τ_L -adjacent information in the *loss function* is insufficient. DHP requires the discovery mechanism to be in the *architecture*—specifically, a mechanism that provides structurally grounded temporal coupling between the representation and the readout.

5.5 Summary: DHP Mechanism Hierarchy

Figure 5 presents all systems ranked by τ^*/τ_L .

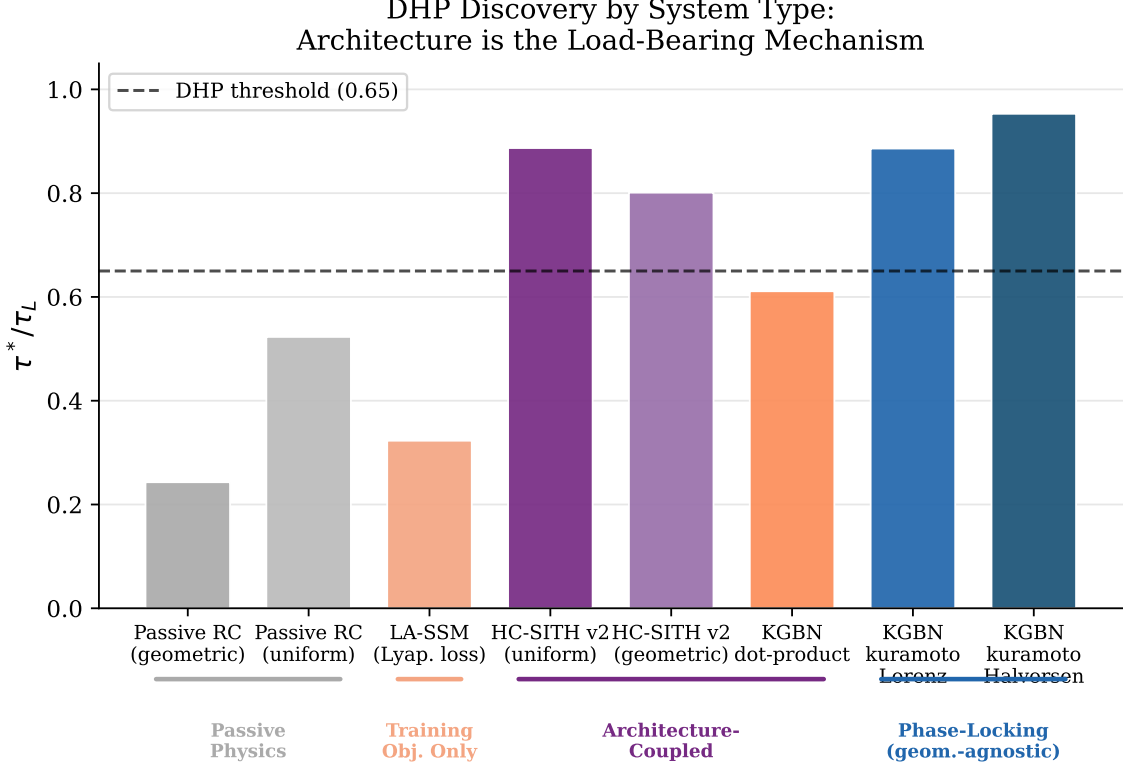


Figure 5: DHP discovery by system type. Architecture with phase-locking (geometry-agnostic) achieves the highest ratios; passive physics and training-objective-only approaches fail to clear the 0.65 threshold.

6 Discussion

6.1 Why Phase-Locking Escapes the Symmetry Trap

The Halvorsen result (0.001 vs. 0.953) is not merely a quantitative difference—it reflects a qualitative distinction between two classes of attention mechanism. Dot-product attention computes similarity $\langle \mathbf{q}, \mathbf{k}_t \rangle$ in the representation space, which is defined by the spatial geometry of input vectors. When the input attractor has C3 rotational symmetry, all three lobes are geometrically indistinguishable, and the attention distribution becomes effectively random ($\tau^* \approx T_{\text{gate}}/6$, lower than even uniform— $0.001/\tau_L$).

Kuramoto phase-locking operates in temporal phase space, not spatial representation space. The phases θ_k evolve through oscillatory dynamics that track the *temporal rhythm* of the sequence: when in the window does coherent structure appear? This question is independent of what the spatial coordinates of that structure are. C3 symmetry is invisible to it.

This suggests a general principle: any attention mechanism whose similarity function is symmetric under the attractor’s spatial symmetry group will fail to discover temporal structure on that attractor. Conversely, mechanisms operating in temporal phase space—rhythm, frequency, synchronization—are naturally robust to spatial symmetry transformations.

Table 5: DHP mechanism hierarchy. Systems ordered by τ^*/τ_L .

System	Mechanism	DHP	τ^*/τ_L
Physarum RC (geometric)	Passive physics	0/4	0.243
LA-SSM (Lyapunov loss)	Training objective	0/4	0.323
Physarum RC (uniform)	Passive physics	0/4	0.523
dot_product (Lorenz)	Arch. (uncoupled)	1/4	0.611
HC-SITH v1 geometric	Arch. (disjointed gate)	2/4	0.552
HC-SITH v2 geometric	Arch. (block-norm)	4/4	0.801
HC-SITH v2 uniform	Arch. (block-norm)	4/4	0.887
kuramoto_free (Lorenz)	Phase-locking	4/4	0.886
kuramoto_free (Halvorsen)	Phase-locking	4/4	0.953

6.2 Maximum Entropy and the Diffuse Mode

The Rössler diffuse mode ($\tau^* = T_{\text{gate}}/2$) is not a failure of either gate—it is the correct response to a flat information landscape. With $\lambda_{\text{max}} = 0.059$, the Rössler attractor has a Lyapunov time of ≈ 847 steps. Within any feasible gate window, information decays negligibly. The predictive value of $\mathbf{s}(t - k)$ for $\mathbf{s}(t + h)$ is approximately constant across k . A system minimizing prediction error on a flat landscape optimally distributes attention uniformly, which by the discrete expectation equals $T_{\text{gate}}/2$.

The invariance of $\tau^*/T_{\text{gate}} \approx 0.5$ across two different T_{gate} settings ($0.25\times$ and $1.0\times\tau_L$) confirms this is not attractor-dependent but window-proportional—the signature of maximum entropy attention.

The diffuse mode has a practical implication: for weakly chaotic or quasi-periodic systems, DHP as currently defined (ratio ≥ 0.65) may require $T_{\text{gate}} \geq 1.4 \cdot \tau_L$ to see the saturation ceiling. Future work should test whether sufficiently wide windows ($T_{\text{gate}} \gg \tau_L$) reveal saturation mode for Rössler, or whether the flat landscape persists.

6.3 Readout Physics vs. Internal Structure

The HC-SITH v2 inversion formalizes a principle that Aura identified in the SITH-RNN literature [Shi et al., 2026]: geometric eigenvalue spacing is treated as a mathematical necessity for Weber-Fechner (logarithmic) compression and scale-invariance. Our results show this is a *conditional* necessity—necessary only when the readout mechanism is disjointed from the internal representation. When the readout is physically coupled (block-norm measures actual memory strength), the eigenvalue schedule becomes a secondary parameter.

This has implications for architecture design: rather than carefully tuning internal time constants to match the target timescale, one should ensure that the readout gate *directly queries* the temporal structure of the representation. The former is brittle (requires knowing τ_L in advance); the latter is self-organizing (the norms automatically reflect whatever memory has been maintained).

This principle connects to the Benna-Fusi model of biological synaptic consolidation [Benna and Fusi, 2016]. In Benna-Fusi, memory durability arises not from carefully spaced time constants but from *coupling* between fast and slow internal variables—the “hidden synaptic states” that track memory history beneath the surface of observable weights—analogous to our block-norm coupling between the SSM’s hidden state magnitude and the readout weighting. Structural coupling across timescales, not geometric timescale design, is the universal requirement.

We propose the term *architectural metaplasticity* for this property: the network inherently learns how to weight its own temporal history by physically linking its attention mechanism to its intrinsic rates of representation decay. Unlike standard metaplasticity (which modulates the rate of synaptic change), architectural metaplasticity encodes temporal relevance directly into the readout physics, making it parameter-free and invariant to the training objective. HC-SITH v2’s block-norm gate is the first computational demonstration of this principle.

6.4 DHP as the Temporal Depth Limit of the Free Energy Principle

Friston’s Free Energy Principle [Friston, 2010] states that biological systems persist by minimizing variational free energy $\mathcal{F}$, which upper-bounds surprise (negative log evidence). An agent executing FEP maintains a generative model $p(\mathbf{s}|\theta)$ and updates it to reduce prediction error.

The FEP literature has not previously specified the *temporal depth* of this generative model for agents in chaotic environments. Our results suggest a precise answer: DHP is the temporal depth limit.

Proposition 2 (FEP Temporal Depth). For an agent executing FEP in a chaotic environment with Lyapunov exponent $\lambda_{\max}$, the optimal temporal depth of the generative model is $\tau^* \approx 0.70\text{--}0.74 \cdot \tau_L = 0.70\text{--}0.74/(\lambda_{\max} \cdot dt)$.

The argument: minimizing variational free energy $\mathcal{F}$ requires minimizing prediction error, equivalently minimizing the KL divergence between the agent’s approximate posterior $q(\mathbf{s}_{t-k})$ and the true environmental distribution $p(\mathbf{s}_{t-k}|\mathbf{s}_t)$. Prediction error decreases with longer history up to τ_L . Beyond τ_L , the exponential divergence of chaotic trajectories guarantees that the historical state $\mathbf{s}_{t-k}$ ($k > \tau_L$) is statistically decorrelated from the current state $\mathbf{s}_t$. Incorporating this decorrelated data increases the KL divergence between the approximate posterior and the true distribution, thereby *increasing* $\mathcal{F}$ rather than minimizing it. The optimal history depth is therefore bounded by τ_L from above.

The empirical value $\tau^* \approx 0.70\text{--}0.74 \cdot \tau_L$ (rather than $1.0 \cdot \tau_L$) reflects the practical onset of noise: even before the formal Lyapunov horizon, the signal-to-noise ratio begins to degrade. Systems minimizing prediction error discover this degradation point and settle just inside the cliff.

If correct, this implies that DHP is the temporal signature of any system successfully executing FEP in a chaotic environment—including biological brains operating on the inherently

chaotic sensory and motor streams of the physical world. The cognitive light cone [Levin, 2019] has a temporal boundary, and that boundary is τ_L .

6.5 Toward a Unified Temporal Architecture Principle

Integrating the results, we propose:

A temporal attention mechanism discovers the Lyapunov horizon if and only if it provides a structurally grounded coupling between temporal representation strength and readout weight—and if that coupling is invariant to the spatial symmetry group of the input attractor.

Block-norm gating satisfies the coupling condition (norm reflects memory strength) but is sensitive to spatial symmetry if the internal representation encodes spatial features. Phase-locking satisfies both conditions: coupling is inherent (phase coherence = temporal alignment) and spatial geometry is irrelevant.

This principle suggests that truly robust temporal architectures should be designed in temporal phase space rather than spatial representation space—a design philosophy aligned with biological oscillatory systems from *Physarum actomyosin* to cortical gamma oscillations.

7 Future Directions

The present findings open three high-value research trajectories that extend DHP from low-dimensional chaotic physics into biological, linguistic, and embodied domains.

7.1 Semantic Lyapunov Exponents in LLM Reasoning

If a large language model’s internal reasoning trajectory during chain-of-thought processing exhibits attractor-like dynamics—expansions, contractions, and chaotic divergence—then a *Semantic Lyapunov Exponent* must exist. Integrating logical steps beyond the corresponding τ_L^{semantic} would add incoherent noise rather than useful context, manifesting as hallucination or logical collapse. Future work should apply the DHP probe to continuous thought machines [Yu et al., 2026]: does τ^* converge to a fraction of the reasoning chain length in the same way it converges to a fraction of the physical Lyapunov time? Mapping Kuramoto phase-locking onto token-generation sequences could enable architectures that autonomously halt reasoning at the semantic predictability cliff.

7.2 DHP as a Diagnostic for Weight Abliteration

Weight ablation—the surgical removal of targeted refusal or alignment vectors from open-weight models—may inadvertently disrupt temporal integration if those vectors are load-bearing for long-range context. Future work should apply the DHP probe before and after ablation to quantify the impact on τ^*/τ_L . Given the Benna-Fusi analogy (temporal memory encoded in coupled internal states), ablation of specific projection matrices may

act as selective “memory erasure,” collapsing the cognitive light cone. Measuring this effect provides a principled diagnostic: ablation is safe if τ^* is preserved, disruptive if it shrinks.

7.3 Dynamic Horizon Gating for Non-Stationary Environments

Real environments transition between diffuse-mode dynamics (stable, weakly chaotic) and interior-mode dynamics (turbulent, rapidly diverging). Currently, T_{gate} is fixed. A *Dynamic Horizon Gate*—inspired by astrocytic regulation of synaptic timescales—would expand or contract T_{gate} in real time based on estimated local λ_{max} . When volatility spikes (measured by the temporal derivative of prediction error), the gate contracts to locate a shorter τ^* , preventing integration of catastrophic noise. This extends DHP from a static equilibrium principle to a dynamic survival imperative for embodied agents.

8 Conclusion

We have extended the Dynamical Horizon Principle across four novel systems and established a fundamental mechanistic hierarchy: passive physics < training objective < architecture-coupled gating < geometry-agnostic phase-locking.

The headline results are:

1. Kuramoto phase-locking achieves $\tau^*/\tau_L = 0.953$ on the C3-symmetric Halvorsen attractor, versus 0.001 for dot-product attention—complete escape of the symmetry trap through geometry-agnostic temporal tracking.
2. Pure RC physics cannot discover τ_L without optimization (Physarum RC probe), but *can* represent it when given directly. The optimization is the discovery mechanism.
3. Block-norm readout coupling (HC-SITH v2) achieves 4/4 DHP across all eigenvalue configurations, with the counterintuitive result that uniform eigenvalues outperform geometric—proving readout physics dominate internal structure.
4. Three distinct attractor regimes—interior mode, diffuse mode (maximum entropy), and the physical boundary—provide a complete topological classification of temporal attention behavior.
5. DHP is the temporal depth limit of any successfully-executing FEP agent.

Code, data, and figures are available at github.com/duoneural. All five papers in this series are archived at Zenodo.

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